通过联合生成与评估实现自我进化的深度研究

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摘要

大语言模型（Large Language Models）在各类日常应用中日益普及，其中深度研究能力尤为关键。与传统问答任务不同，深度研究报告生成缺乏确定性的标准答案，这使得奖励设计本质上难以验证，从而限制了有效的强化学习。现有方法通过大模型作为评审（LLM-as-a-Judge）和依赖查询的评估准则来缓解这一挑战，但它们仍依赖静态评估器，无法随着求解器的改进而调整标准，导致优化压力不足并最终饱和。本文提出一种用于深度研究评估与生成的自我进化协同训练框架（SCORE），该框架在共享参数的学习过程中紧密耦合评估器与求解器。本文并未将生成与评估视为独立模块，而是利用它们的内在联系，在单一共享参数模型中实现联合改进。为约束这一过程，本文引入元控制机制，根据求解器性能动态调控评估环境，鼓励有效的评估维度并确保评估器搜索足够深入。在深度研究基准上的大量实验表明，报告生成质量持续提升，证明评估与生成的协同进化是训练开放式研究智能体的一个有前景的方向。

1 引言

凭借其卓越的能力，大语言模型（Large Language Models）已深度融入众多复杂领域，涵盖自动驾驶 [48]、软件开发 [12]、多媒体生成 [33, 46, 54] 等 [49, 26]。随着对全面信息检索与综合需求的增长，提升大语言模型的深度研究能力日益受到研究者的关注。若干搜索智能体，如 GPT-Researcher [1] 和 AgentCPM-Explore [5]，近期已在多跳研究任务中展现出卓越的能力。此外，先前的工作如 Search-o1 [27] 和 SSP [31] 表明，将强化学习（Reinforcement Learning）应用于这些模型并集成到智能体中，能够显著增强其推理能力，在复杂问答任务上取得显著的性能提升。

与传统的问答任务不同，训练开放式报告生成模型面临两大挑战。首先，综合性报告缺乏明确的基准真值，使得客观奖励信号本质上难以定义 [24, 15]。由于开放式查询没有绝对标准，依赖人工标注来评估生成报告的成本极其高昂，导致客观且可扩展的奖励信号难以构建。依赖人工标注来覆盖这些多样化的路径成本极高且难以扩展。其次，评估标准 * 同等贡献； † 通讯作者。

预印本。

高质量报告的评价标准是多维且与查询相关的 [22, 13]。单一固定的评估指标不可避免地忽略了关键维度，包括不同问题的事实正确性和洞见多样性。这种维度缺失导致训练信号稀疏，无法为复杂生成任务提供所需的细粒度贡献分配。

为了克服报告生成中的这些挑战，近期研究提出了几种新颖的范式，以提供更有效的优化信号。WebThinker [28] 和 Step-DeepResearch [18] 已将大模型作为评审（LLM-as-a-judge）[57] 应用于训练过程中评估报告质量。AdaRubric [11] 提出了一个框架，能够根据输入指令动态生成任务特定的评估标准。Mix-GRM [52] 利用带可验证奖励的强化学习（RLVR），根据不同的任务需求自动在广度优先（B-CoT）和深度优先（D-CoT）评估策略之间进行适配。然而，固定的外部评估器缺乏动态调整其评估标准的能力，因此无法在求解器改进时提供持续的优化压力。

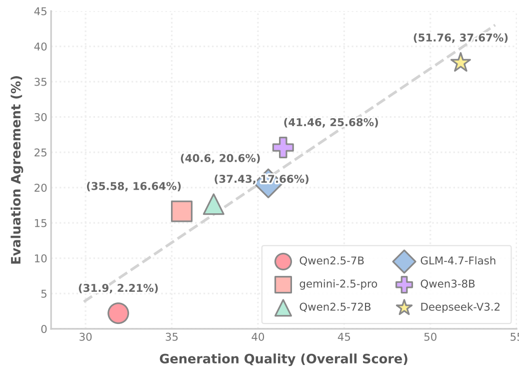


图 1：大语言模型生成能力与评估能力之间正相关性的实证证据。

如图 1 所示，经验观察表明，大语言模型的生成质量与其评估能力呈正相关。基于这一洞见，我们假设生成与评估能力可以相互促进，并应在统一的而非解耦的架构中进行优化。为此，我们提出一种自我博弈框架，其中策略模型迭代地生成和评估报告。通过用随求解器共同演化的动态评估器取代固定的外部评判器，我们的方法在整个学习过程中保持了有效的训练压力。此外，应用元约束框架 [23] 通过施加环境约束来稳定这一过程。实验表明，这种共同演化范式在多种研究生成任务上带来了显著且一致的改进。

简而言之，我们的贡献总结如下：

- 我们提出了 SCORE，一种用于训练具有不可验证奖励的深度研究智能体的评估器-求解器自我博弈框架。该框架提供动态、多维度的奖励，以优化求解器在开放式报告生成中的能力，并受元约束框架的约束。
- 我们在共享参数下对 SCORE 进行了理论分析。我们的分析阐明了评估器侧自适应中一致性的作用，并刻画了共享参数交替更新的局部优化行为。
- 我们展示了我们的方法在深度研究基准上多个评估维度上提升了智能体性能，并对不可验证奖励下参数如何影响训练过程进行了系统分析。

2 相关工作

2.1 深度研究

在近期的智能体研究中，大语言模型（LLMs）已展现出强大的深度研究能力 [59, 4]。早期框架，包括 Search-o1[27]、Search-R1[21] 和 R1-Searcher[39]，成功地将显式推理过程引入智能体搜索框架，在多跳问答（QA）任务中取得了显著的实证改进。在这些进展之后，研究者将多种优化算法集成到智能体训练流程中。具体而言，WebThinker[28] 利用迭代直接偏好优化（DPO）训练深度研究智能体，而 DeepResearcher[58] 则采用群体相对策略优化（GRPO）。与这些算法增强不同，SimpleDeepSearcher[40] 认为训练数据的质量比特定的训练范式更为重要。除了问答任务之外，报告生成任务本质上更为复杂，因为它们

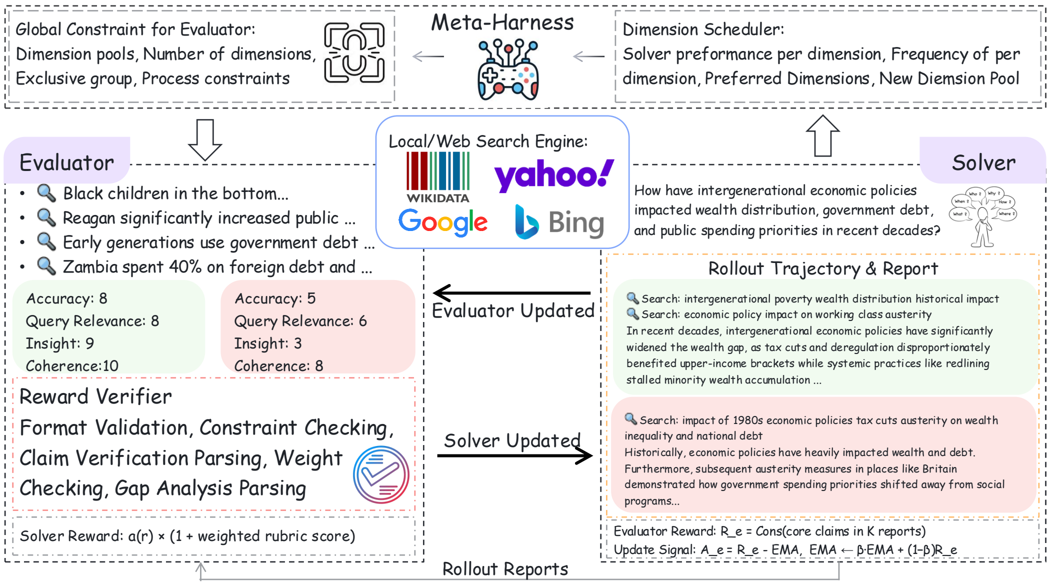


图 2：SCORE 由元控制器（Meta-Harness）控制。评估器首先从提供的维度池中选择评估维度，并据此检索证据。求解器利用来自评估器的奖励进行更新，而评估器则基于求解器生成的展开报告的一致性进行优化。奖励验证器在奖励用于求解器优化之前过滤掉无效奖励。

涉及开放式、多方面的综合，其中为自动评估建立可靠的真实标签仍然是一个关键瓶颈。为了应对这种复杂性，像 STORM 和 Co-STORM 这样的多智能体框架在报告生成过程中利用先验知识和人在回路交互 [34, 20]。后来的方法，如 Step-DeepResearch 和 AgentCPM-Report[18, 29]，直接集成了针对其特定智能体框架优化的模型训练。

2.2 内在奖励与自我博弈

传统自我博弈（Self-play）作为一种对抗性强化学习方法，被广泛应用于智能体研究 [53]。AbsoluteZero、R-Zero、SeRL 和 SPIRAL 等工作证明了在数据稀缺情况下，自我进化（self-evolution）在解决复杂推理任务中的有效性 [55, 19, 30, 14]。近期框架，包括 Search Self-play、Dr. Zero 和 SPaR，展示了将该技术应用于多跳环境中搜索智能体的显著优势 [31, 50, 6]。除了提出者-求解者自我博弈等对抗性学习范式外，替代方法提出在模型自身生成的过滤轨迹上训练求解器，如 WebThinker、EvolveSearch 和 EXSEARCH 等框架所示 [28, 51, 38]。其他值得注意的工作包括 AceSearcher，它利用分解器与求解器之间的协作交互实现自我改进 [45]。此外，利用模型自身提供的内部奖励的优化策略已被 INTUITOR [56] 和 JEPO [41] 等框架的研究者采用。这些先前工作证明了内部信号在增强自主能力改进方面的有效性。然而，深度研究任务的独特特性带来了显著的评估挑战。对开放域、综合性报告进行可靠评分的困难，使得设计能够解决信用分配问题的有效奖励信号极具挑战性。

3 方法论

3.1 问题设定

本文研究深度研究报告生成，其目标是在给定查询 q 和环境 $\mathcal{E}_q$ 的情况下，学习候选报告 r 上的策略 $\pi(r \mid q, \mathcal{E}_q)$ 。假设报告质量由潜在函数 $Q(r; q, \mathcal{E}_q)$，决定，理想目标是为潜在质量更高的报告分配更多概率质量

此处， $q \in \mathcal{Q}$ 表示用户查询， $r \in \mathcal{R}$ 表示候选长文报告， $\mathcal{E}_q$ 表示查询特定的证据与评估环境。

与问答式任务不同， Q 本质上是多维且依赖于查询的：一份报告可能在不同维度上成功或失败，例如事实支持、证据覆盖、引用依据、矛盾处理、不确定性校准和论述连贯性。因此，长文报告生成很少存在单一可靠的验证器或固定标量奖励，既能提供足够信息又具有广泛适用性。

因此，本文将深度研究形式化为一个以查询为条件、以评估为瓶颈的报告优化问题。具体而言，训练深度研究模型的核心挑战在于，如何在有限监督下，以查询特定且稳定的方式评估长篇幅报告。

3.2 SCORE 概述

为应对这一挑战，本文提出 SCORE 框架，一种面向深度研究报告生成的检索增强与外部引导的评估求解器框架，如图 2 所示。SCORE 包含四个组件：

> 1. 一个可通过搜索与检索工具进行查询的外部证据环境，2. 一个固定的外部元框架，用于塑造特定查询的评估环境，3. 一个评估器，构建以查询为条件的评分标准并生成结构化报告评估，4. 一个求解器，与证据环境交互，收集支持性证据，并在所得评分标准下生成报告。

SCORE 的一个关键设计选择是，求解器和评估器在共享的底层执行器上实现为两个功能不同的角色。这一设计的动机在于报告生成与报告评估之间存在大量重叠，两者都需要查询理解、证据建模和篇章级推理。我们在消融实验第 5.2 节中对该选择进行了实证检验。

训练通过共享行动者的顺序交替更新进行：求解器利用评估者衍生的报告奖励进行优化，而评估者则利用评分标准级别的辅助信号进行优化。两种更新均通过针对参考策略的 KL 散度进行正则化，以限制过度的策略漂移。

3.3 外部证据与查询特定环境

对于每个查询 q ，SCORE 通过搜索和检索工具与外部证据环境进行交互。我们用

$$D_q^{(0)} = \mathcal{R}(q)$$

从该环境获得的初始证据，其中 $\mathcal{R}$ 抽象了系统可用的检索交互。在实践中，该证据环境可能包括网络搜索、文档检索或其他工具介导的证据来源。

基于初始证据 $D_q^{(0)}$ ，SCORE 构建了一个查询特定的评估环境 通过一个固定的外部元控制器：

$$\mathcal{E}_q = \mathrm{Harness}(q, D_q^{(0)}, \mathcal{S}),$$

其中 $\mathcal{S}$ 表示控制器可用的近期训练统计或历史信号。由此产生的环境 $\mathcal{E}_q$ 指定了当前查询的可接受评估空间，包括候选评估维度、有效评分标准的结构约束、证据验证或比较的过程要求，以及辅助的任务特定指导。

元控制器是外部的且不可训练的。它被实现为一个固定的外部控制器，不通过梯度下降与参与者一起优化。其作用是塑造评估者-求解者交互发生的可接受评估空间。

3.4 评估者与求解者

给定查询 q 和环境 $\mathcal{E}_q$，评估者采样一个查询条件的评分标准

$$(H_q, w_q) \sim \pi_e(\cdot \mid q, \mathcal{E}_q),$$

其中 $H_q = \{h_1, \ldots, h_k\}$ 表示选定的评估维度， w_q 表示它们对应的权重。

在给定查询、当前证据状态以及评估器所选评分标准（Rubric）的条件下，求解器与外部证据环境交互，收集支持信息并生成一组候选报告 $r_1, \ldots, r_K \sim \pi_s(\cdot \mid q, D_q^s, H_q, w_q),$，其中 D_q^s 表示求解器收集到的证据， K 是训练期间针对同一查询采样的求解器展开次数。

在评估阶段，评估器可基于查询和已生成的报告进行受限的补充检索或验证，从而构建自身的证据上下文 D_q^e。随后，评估器对候选报告生成结构化判断，包括分维度评分、证据支持检查、矛盾分析以及引用验证。这些结构化判断被转换为求解器侧的学习信号。

3.5 共享行动者与训练信号

SCORE 将求解器和评估器实现为共享行动者（Actor）上的两种角色，其参数为 θ。这一设计源于以下观察：报告生成与报告评估依赖于重叠的潜在能力，包括查询理解、证据建模和篇章级推理。基于这一视角，生成与评估不被视为完全独立的功能，而是从共享底层表示中诱导出的两种角色特定映射。因此，共享行动者为两种角色提供了共同的表示基础，使评估器侧的适应能够影响求解器所使用的相同内部表示。我们在第 5.2 节对这一设计选择进行了实证检验。

SCORE uses two distinct reward channels:

$$\underbrace{R_s(r; q, D_q^e, H_q, w_q)}_{\text{solver-side report reward}} \quad \text{and} \quad \underbrace{R_e(\Gamma_q) = C(\Gamma_q)}_{\text{evaluator-side auxiliary reward}}.$$

求解器侧奖励监督报告生成，而评估器侧奖励则调整评估器的评分标准选择行为。

求解器侧报告奖励。 对于每份生成的报告 r，评估器产生一个标量报告奖励 $R_s(r; q, D_q^e, H_q, w_q),$，其中 D_q^e 表示评估器用于基于评分标准的条件评估的证据上下文， $H_q = \{h_1, \ldots, h_m\}$ 表示选定的评估维度， $w_q = \{w_1, \ldots, w_m\}$ 表示其对应权重。该奖励作为优化求解器的主要监督信号。

在我们的实现中，抽象的报告奖励被具体化为

$$R_s(r) = a(r) \cdot \left(1 + \sum_{j=1}^{m} w_j \, s_j(r; q, D_q^e)\right),$$

其中 $a(r)$ 表示报告级别的有效性或准确率得分， $s_j(r; q, D_q^e)$ 表示在评估者侧证据上下文 D_q^e 下，报告 r 在维度 h_j 上的评估者得分。直观上，求解器被鼓励生成既有效又在评估者所选评分维度上表现强劲的报告。

评估者侧辅助奖励。为了适应评估者，SCORE 引入了基于报告间一致性的评分维度级辅助奖励。对于固定的查询和评估者所选的评分维度

$$\Gamma_q := (H_q, w_q),$$

求解器在相同的查询和评估设置下生成 K 份报告。我们将原始评估者侧辅助奖励定义为

$$R_e(\Gamma_q) = C(\Gamma_q),$$

其中 $C(\Gamma_q)$ 是通过聚合生成报告之间的成对一致性计算得到的经验一致性得分。在当前实现中，$C(\Gamma_q)$ 被计算为提取的报告结论上成对一致性得分的批次均值。

评估者更新使用该信号的以指数移动平均为中心的形式，

$$A_e(\Gamma_q) = R_e(\Gamma_q) - b_{t-1},$$

其中 b_{t-1} 是过去辅助奖励的先前指数移动平均基线。在计算中心化信号后，基线更新为

$$b_t = \beta b_{t-1} + (1-\beta) R_e(\Gamma_q),$$

其中 β 是指数移动平均衰减系数。因此，R_e 偏好那些在重复采样中求解器行为更具可重复性的评分维度，而 A_e 提供了用于强化式评估者更新的实际低方差信号。

总体而言，SCORE 形成了一个双通道训练循环：评估器衍生的报告奖励改进求解器，而基于一致性的评分标准级奖励调整评估器。求解器侧奖励针对所选评分标准下的报告质量，而评估器侧奖励则偏好能够诱导更稳定且可复现的求解器行为的评分标准。在此意义上，一致性被用作面向稳定性的辅助信号，用于评分标准自适应，而非直接的正确性判据。

3.6 顺序交替 KL 正则化优化

鉴于第 3.5 节定义的两个奖励通道，SCORE 中的训练通过对共享行动者的顺序交替更新进行。每一步构建一个查询特定的评估环境，生成以评分标准为条件的报告，在评估器侧验证上下文中对其进行评估，然后对共享参数应用求解器侧和评估器侧更新 θ 。

求解器侧更新使用评估器衍生的报告奖励 R_s ，将其转换为组归一化优势，并通过 GRPO 进行优化。评估器侧更新使用以 EMA 为中心的辅助奖励 A_e （源自原始一致性奖励 R_e ），以 REINFORCE 风格的目标更新评分标准选择策略。尽管这些更新是从不同的轨迹和奖励中计算得出的，但两者都作用于同一个行动者，因此通过共享参数化相互耦合。

两种更新均通过相对于固定参考策略 π_{ref} 的 KL 散度进行正则化。在我们的实现中，该 KL 项在行动者更新层面应用，而非作为显式奖励惩罚。因此，KL 正则化不被视为第三个奖励通道，而是作为一种优化正则化器，限制共享行动者学习下的过度策略漂移。

在分析层面，我们将未正则化的角色特定目标分别记为 $\mathcal{L}_s(\theta)$ 和 $\mathcal{L}_e(\theta),$。相应的 KL 正则化目标可写为

$$\mathcal{L}_s^{\text{reg}}(\theta) = \mathcal{L}_s(\theta) + \lambda\,\text{KL}(\pi_\theta \,\|\, \pi_{\text{ref}}),$$

$$\mathcal{L}_e^{\text{reg}}(\theta) = \mathcal{L}_e(\theta) + \lambda\,\text{KL}(\pi_\theta \,\|\, \pi_{\text{ref}}),$$

其中 $\lambda > 0$ 是 KL 正则化系数。

KL 正则化在 SCORE 中具有两个作用。首先，它通过在共享角色学习下限制策略的突然漂移来稳定交替优化。其次，它有助于在共享行动者上保持求解器侧和评估器侧更新之间的兼容性。通过这种方式，SCORE 将评估器衍生的报告奖励、基于一致性的评估器自适应、外部基础、共享角色学习以及 KL 正则化的交替优化结合到一个统一的训练框架中，用于生成长篇报告。

4 理论分析

我们分析了 SCORE 的两个性质：一致性作为评估器侧辅助信号的作用，以及共享参数 KL 正则化交替更新的一阶优化行为。

假设。我们假设：（A1）分析级目标 $\mathcal{L}_s$ 和 $\mathcal{L}_e$ 是 L -光滑的；（A2）相应的随机梯度是无偏的或近似无偏的，且方差有界；（A3）随机梯度几乎必然有界。

一致性作为由评分标准诱导的可重复性。令 $\Gamma_q := (H_q, w_q)$ 表示评估器为查询 q 选择的评分标准。假设求解器在相同的查询、证据上下文和评分标准下生成 K 份报告，并令 $C(\Gamma_q)$ 表示通过聚合这些报告之间的成对一致性得分而得到的经验一致性分数。在当前实现中，这是生成报告之间的平均成对一致性。我们将原始评估器侧辅助信号定义为 $R_e(\Gamma_q) = C(\Gamma_q),$，并使用中心化信号 $A_e(\Gamma_q) = R_e(\Gamma_q) - b_t$ 进行评估器更新，其中 b_t 是指数移动平均基线。

命题 1（一致性估计由评分标准诱导的可重复性）。令 $\mu_C(\Gamma_q)$ 表示在评分标准 Γ_q 下一致性度量的总体均值。则 $\mathbb{E}[R_e(\Gamma_q)] = \mathbb{E}[\mu_C(\Gamma_q)].$ 。因此，原始评估器侧辅助信号估计了在重复求解器采样下由评分标准 Γ_q 诱导的期望可重复性。

更高的可重复性表明，该评分标准为求解器引入了更稳定且可重复的评估环境，使其成为评估器端自适应的有效辅助信号。以指数移动平均基线为中心化处理，在不改变原始信号所编码的潜在偏好的前提下，降低了梯度方差。因此，在 SCORE 中，一致性被用作评估器端自适应的评分标准级辅助信号。

交替更新的一阶刻画。设共享智能体的更新为 **ion of alternating** **tes.** Let the shared-a

$$\theta' = \theta_t - \eta_s \hat{g}_s(\theta_t), \qquad \theta_{t+1} = \theta' - \eta_e \hat{g}_e(\theta'),$$

其中 $\hat{g}_s$ 和 $\hat{g}_e$ 分别是求解器端和评估器端目标的随机梯度。定义分析代理

$$\widetilde{\mathcal{L}}(\theta) = \mathcal{L}_s(\theta) + \alpha \mathcal{L}_e(\theta) + \lambda \, \mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}), \qquad \alpha = \eta_e / \eta_s,$$

其中 π_{ref} 是固定参考策略， $\lambda > 0$ 是 KL 正则化系数。

命题 2（交替更新的一阶刻画）。在假设（A1）–（A3）下，期望的单步 SCORE 更新满足

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \widetilde{\mathcal{L}}(\theta_t) + \rho_t, \qquad \|\rho_t\| = O(\eta_s^2).$$

该结果表明，共享参数的求解器-评估器更新在一阶近似下遵循 KL 正则化的代理下降方向，仅由顺序交替更新引入高阶残差。详细推导见附录 F。

5 实验

5.1 实验设置

训练细节 本文使用 VeRL 框架 [36] 训练所有模型。评估器与求解器异步更新：求解器在每个训练步使用 GRPO [35] 更新，而评估器每 1 步使用 REINFORCE [44] 更新一次。评估器的 rollout 数设为 1，求解器为 8。最大工具调用次数为 10，批大小为 64，最大响应长度为 4,096 个 token，最大输入长度为 8,192 个 token。训练时使用基于本地 WikiData 的本地搜索引擎，评估时使用网络搜索。GPT-5.2 [3] 作为元工具（Meta-Harness），每 5 个训练步调用一次，根据求解器当前性能修改环境。对于 EMA 基线的系数，本文设 β =0.7。训练集通过收集并预处理 Reddit 用户查询构建。

表 1： 集成 LLaMA-3.1 和 Qwen2.5 的 open-deep-research 与 gpt-researcher 在 DeepResearchBench 上的性能。Comp.：全面性，**IF**：指令遵循，Read.：可读性，**VC**：引用有效比例。

Models and Agents	**Writing Quality**					**VC.**
	Overall	**Comp.**	**Insight**	**IF.**	**Read.**	
Agent: Open-deep-research (ReAct Paradigm)						
Llama-3.1-8B-Inst.	27.00	24.05	22.46	30.28	36.29	22.58
+ GRPO	23.41 (-3.59)	19.89 (-4.16)	18.73 (-3.73)	26.77 (-3.51)	33.90 (-2.39)	28.57 (+5.99)
+ DPO	26.13 (-0.87)	23.11 (-0.94)	21.28 (-1.18)	29.84 (-0.44)	35.77 (-0.52)	28.00 (+5.42)
+ SCORE (Ours)	30.07 (+3.07)	30.31 (+6.26)	26.71 (+4.25)	32.25 (+1.97)	33.66 (-2.63)	30.55 (+7.97)
Qwen2.5-7B-Inst.	31.92	28.93	27.81	35.37	40.29	11.76
+ GRPO	32.67 (+0.75)	29.94 (+1.01)	29.08 (+1.27)	35.30 (-0.07)	40.87 (+0.58)	16.13 (+4.37)
+ DPO	33.91 (+1.99)	34.26 (+5.33)	30.89 (+3.08)	35.90 (+0.53)	36.85 (-3.44)	13.76 (+2.00)
+ SCORE (Ours)	34.43 (+2.51)	34.86 (+5.93)	31.38 (+3.57)	36.37 (+1.00)	37.41 (-2.88)	19.35 (+7.59)
Agent: gpt-researcher (Plan-and-Execute Paradigm)						
Llama-3.1-8B-Inst.	28.92	25.62	24.80	31.85	38.36	38.23
+ GRPO	16.97 (-11.95)	16.79 (-38.23)	13.12 (-11.68)	18.09 (-13.76)	23.78 (-14.58)	0.00 (-8.16)
+ DPO	33.58 (+4.66)	33.83 (+8.21)	30.64 (+5.84)	35.15 (+3.30)	36.87 (-1.49)	34.48 (-3.75)
+ SCORE (Ours)	33.89 (+4.97)	34.14 (+8.52)	30.90 (+6.10)	35.64 (+3.79)	37.29 (-1.07)	52.38 (+14.15)
Qwen2.5-7B-Inst.	32.00	29.20	28.70	34.67	39.61	36.11
+ GRPO	30.03 (-1.97)	27.52 (-1.68)	27.39 (-1.31)	31.07 (-3.60)	38.37 (-1.24)	51.61 (+15.50)
+ DPO	35.85 (+3.85)	36.22 (+7.02)	33.64 (+4.94)	37.15 (+2.48)	38.07 (-1.54)	30.80 (-5.31)
+ SCORE (Ours)	35.91 (+3.91)	36.15 (+6.95)	33.12 (+4.42)	37.60 (+2.93)	38.93 (-0.68)	66.15 (+30.04)

基线与模型 本文在两个强大的基础大语言模型 Qwen2.5-7B-Instruct [32] 和 Llama-3.1-8B-Instruct [16] 之上实现 SCORE。为进行比较，本文使用标准 GRPO 和 DPO 训练这些模型作为基线。后训练模型随后在两个代表性搜索智能体系统中进行评估，以考察其下游能力。具体而言，本文采用 open-deep-research [2] 作为 ReAct 范式的基线，采用 gpt-researcher [1] 作为计划-执行范式的基线。为保持实验一致性，在相应智能体的所有功能模块中部署同一个统一模型。

基准与评估度量 为严格评估本文方法，我们采用两个专为深度研究任务设计的基准，即深度研究基准（DeepResearchBench）[13] 与深度研究评估（DeepResearchEval）[43]。深度研究基准从全面性、洞见、指令遵循、可读性及引用有效性等多个维度评估智能体性能。相比之下，深度研究评估采用动态评估度量，这与本文的方法论视角高度一致。两个框架最终均采用 GPT-5.2 [57] 进行最终评估。

5.2 实验结果

主要结果 如表 1 所示，SCORE 在所有智能体上均展现出分析深度与有效引用方面的显著且稳健的提升。基础强化学习方法如 GRPO 与 DPO 在至少一个维度上遭受灾难性退化，在集成 LLama-3.1-8B-Instruct 的规划执行智能体中偶尔会完全降至零。这一现象由惰性似然位移（LLD）驱动，即正确推理轨迹的似然在工具集成强化学习过程中衰减，最终导致训练崩溃 [10]。此外，智能体架构放大了这种不稳定性，与逐步推理策略相比，规划执行这类解耦的长时程框架表现出过早崩溃与严重的奖励黑客行为 [9]。相比之下，SCORE 通过利用协同进化的评估器，借助动态评估准则提供密集且一致的约束，从而保持稳定性能。值得注意的是，我们观察到几乎所有训练方法的可读性均出现一致下降。我们假设这是一种根本性的权衡，因为模型被优化以综合全面分析并严谨引用证据，生成的报告固有地包含更密集的学术词汇

以及更复杂的逻辑结构，从而牺牲对话简洁性以换取分析严谨性，这可能类似于多样性与事实性之间的权衡 [25]。

模块消融实验 参照图 3，我们观察到移除任一单独模块都会严重削弱大语言模型的全面性，表明生成全面的研究报告需要整个框架的协同运作。冻结求解器会导致引用有效性大幅下降。评估器的内部一致性奖励带来了奖励黑客行为，即评估器学会构建肤浅的评分标准，从静态求解器中诱导出高度一致的文本，同时主动规避诸如严格外部引用等高方差行为。此外，与消融基线相比，我们的完整方法在可读性得分上略低，我们将其归因于自然语言连贯性与学术格式之间的固有权衡。基础模型倾向于生成流畅但结构松散的文本，而我们的协同进化框架迫使模型生成具有复杂引用语法的更深入报告，从而牺牲表面可读性以换取事实严谨性。

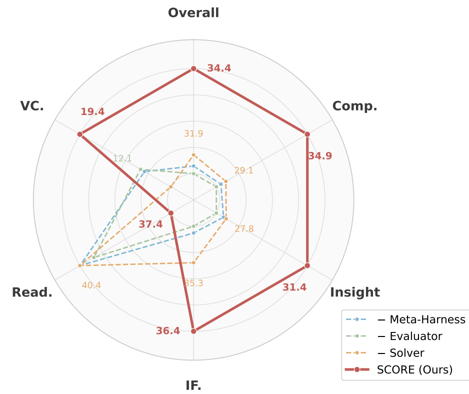
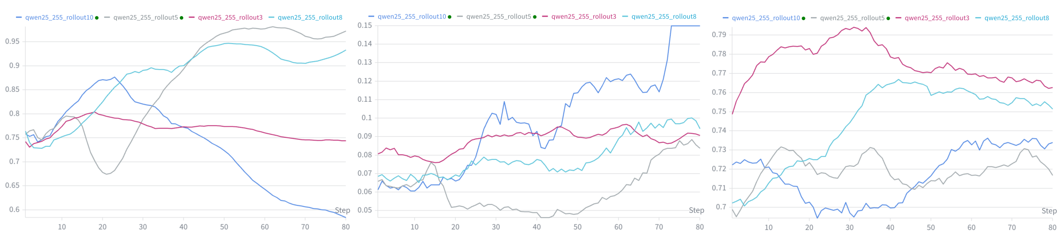


图 3：深度研究基准上各模块的消融实验。 全面：全面性，指令遵循：指令遵循，可读：可读性，引用有效：引用有效比率。



(a) Solver Entropy　(b) Evaluator Reward　(c) Solver Reward

图 4：训练细节——关于展开次数的实验 我们研究了训练过程中展开次数 K 的影响。 图 4 表明，极端的 K 会影响训练过程：较小的 $K = 3$ 无法带来有意义的奖励提升，而较大的 $K = 10$ 会引发严重的模型崩溃，表现为熵 [37] 的突然下降。较小的 K 由于排序样本不足，无法为求解器提供统计上稳健的奖励；而对于评估器，较少的展开次数会产生不可靠的成对一致性估计，导致噪声 REINFORCE 梯度，从而阻碍维度策略的稳定更新。相反，过大的 K 会为求解器和评估器产生高度精确的优势估计，加速它们的更新。在自进化设置下，这种相互加速会加剧策略漂移，并可能引发快速的模型崩溃。

案例研究 如附录 G.2 中的图 6 所示，不同类型的科研查询本质上需要不同的评估标准。例如，法律查询要求绝对的精确率和对法规文本的严格引用，而开放式问题则需要聚合多元视角以保持公平性与客观性。通用的静态评分标准无法捕捉这些关键细微差别，往往会给表面流畅但根本上有缺陷的报告打出可接受的分数。相比之下，本文提出的自适应评估框架动态引入针对每个查询量身定制的专门评估维度。该机制确保生成的报告根据任务所要求的具体严谨性得到准确的扣分或加分。

6 结论

为深度研究生成全面报告是一项高度复杂的任务，而评估这些开放式报告的内在困难严重阻碍了训练过程。我们观察到

模型评估能力与生成能力之间存在强正相关性。基于这一洞见，本文提出 SCORE，一个自进化训练框架，联合优化评估器与求解器。由于整个训练过程从根本上依赖于不可验证的奖励，本文引入外部元约束，对评估器施加必要的环境约束。该模块同时引导评估器在训练阶段主动探索更广泛的相关评估维度。最终，实验结果表明，本文方法在仅需极少训练数据的情况下，实现了显著的性能提升。

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A　局限性

尽管我们的自进化框架具有有效性，但我们承认当前研究中存在若干局限性。首先，深度研究报告生成的独特性质要求基础语言模型具备卓越的指令遵循能力。基础能力较弱的模型可能在动态评分标准生成和工具调用方面遇到困难。其次，我们的训练范式严重依赖于完全不可验证的奖励。缺乏真实标签使得模型在优化过程中容易发生策略崩溃。因此，需要仔细调整超参数并采用严格的早停机制以确保训练稳定性。尽管我们的框架成功为特定标准提供了差异化的奖励信号，但激进地优化某一维度有时会导致另一维度的性能下降。同时平衡这些相互竞争的目标仍然是一个具有挑战性的问题，有待未来探索。

B 未来工作

根据我们的实验，我们观察到在深度研究报告生成的训练过程中仍然存在固有的权衡。我们假设这一问题主要是由极长的智能体轨迹导致的稀疏奖励信号引起的。为了应对这一挑战，未来的工作将探索分阶段训练范式。通过将深度研究生成过程分解为不同的功能阶段并分配中间奖励，我们旨在解决长期困扰端到端策略优化的信用分配问题。

C 大语言模型使用声明

大语言模型被用于提升论文的语法和语言质量。大语言模型也被用于检查证明的正确性和清晰度。在实验中，开源大语言模型被用于后训练模型，而闭源大语言模型被用作评估的裁判。此外，一个编码智能体被用于协助开发训练框架的部分内容。

D 许可证

我们在实验中使用了以下模型：

- Qwen2.5-7B-Instruct、Qwen2.5-72B-Instruct [32] 和 Qwen3-8B [47]，许可证为 Apache 2.0。
- Llama-3.1-8B-Instruct [16]，许可证为 Llama 3.1 Community。
- GLM-4.7-Flash [42] 和 DeepSeek-V3.2 [8]，许可证为 mit。
- Gemini-2.5-pro [7]，许可证为 Apache 2.0；GPT-5.2 [3]，许可证为专有。

E 算法

算法 1 总结了主要的共享参与者评估器-求解器训练循环，而算法 2 详细说明了评估器侧一致性奖励构建和评分标准自适应步骤。在当前实现中，成对一致性被实例化为提取的展开结论上的平均成对 TF-IDF 余弦相似度。

Algorithm 1 SCORE: Evaluator–Solver Self-Play with Consistency-Driven Dimension Learning

Require: Shared actor parameters θ; evaluator role policy $\pi_e(\cdot;\theta)$; solver role policy $\pi_s(\cdot;\theta)$; dimension pool $\mathcal{D}$; solver rollout size K; batch size B; evaluator update period U; harness refresh period M; minimum verification/search depth τ_s; KL coefficient λ_{KL}
Ensure: Trained shared actor parameters θ

Initialize DimensionScheduler $\mathcal{S}$ over $\mathcal{D}$
Initialize Meta-Harness state $\mathcal{M}_0 \leftarrow \mathcal{M}_{\text{default}}$
Initialize evaluator-side EMA baseline $b_0 \leftarrow 0$
for each training step $t = 1, 2, \ldots, T_{\max}$ **do**
  ▷ *Phase 1: Refresh scheduler and harness constraints*
  $(\mathcal{R}_t, \mathcal{A}_t) \leftarrow \mathcal{S}.\textsc{GetConstraints}()$ ▷ retrieve the current required / avoided dimensions and exploration hints
  Obtain current dimension and process constraints from $\mathcal{M}_{t-1}$ ▷ the meta-harness defines the admissible rubric space for this step
  ▷ *Phase 2: Evaluator samples a rubric*
  Sample query batch $\{q_i\}_{i=1}^{B}$
  Build evaluator rubric-generation prompt x_t^{rubric} from $(\mathcal{R}_t, \mathcal{A}_t, \mathcal{M}_{t-1})$
  Sample rubric

$$(H_t, w_t),\ \log \pi_e(H_t, w_t \mid x_t^{\text{rubric}}; \theta) \sim \pi_e(\cdot \mid x_t^{\text{rubric}}; \theta)$$

  ▷ the evaluator chooses what dimensions to emphasize and how to weight them
  ▷ *Phase 3: Solver performs rubric-conditioned rollouts*
  for each query q_i, each rollout $k = 1, \ldots, K$ **do**
    Generate report trajectory and final report

$$y_i^k \sim \pi_s(\cdot \mid q_i, H_t, w_t; \theta)$$

    Record solver-side evidence / tool observations $D^s_{q_i,k}$ ▷ the solver searches, interacts with tools, and produces a long-form report
  end for
  ▷ *Phase 4: Evaluator assesses reports*
  Optionally build evaluator-side evidence contexts $\{D^e_{q_i}\}_{i=1}^{B}$ through constrained retrieval / verification ▷ the evaluator may gather its own evidence for rubric-conditioned assessment
  for each query q_i, each rollout $k = 1, \ldots, K$ **do**
    Compute evaluator-side report reward

$$s_i^k \leftarrow \textsc{EvalScore}(q_i, y_i^k, H_t, w_t, D^e_{q_i}, \mathcal{M}_{t-1}; \theta)$$

    if evaluator verification/search depth for (i, k) is below τ_s **then** set $s_i^k \leftarrow 0$ ▷ discard under-verified rewards to avoid noisy solver supervision
  end for
  ▷ *Phase 5: Solver-side GRPO update*
  for each query q_i **do**
    $\mu_i \leftarrow \text{mean}_k\, s_i^k, \quad \sigma_i \leftarrow \text{std}_k\, s_i^k$
    $\hat{A}_i^k \leftarrow (s_i^k - \mu_i)/(\sigma_i + \varepsilon)$ ▷ group-normalized rewards define solver-side advantages
  end for
  Update shared actor parameters θ using the solver-side GRPO objective with KL regularization against π_{ref}
  ▷ *Phase 6: Evaluator-side consistency update*
  Compute the evaluator-side consistency signal and update the evaluator role as detailed in Algorithm 2 ▷ the evaluator is rewarded when its rubric induces more reproducible solver behavior
  ▷ *Phase 7: Refresh scheduler and meta-harness*
  $\mathcal{S}.\textsc{UpdateStats}(H_t, \{C_i\}_{i=1}^{B})$ ▷ record which dimensions produced more reproducible rollouts
  if $t \bmod M = 0$ **then**
    Collect harness statistics $\mathcal{T}_t$
    $\mathcal{M}_t \leftarrow \textsc{Harness.Generate}(\mathcal{T}_t, \mathcal{M}_{t-1}, \mathcal{M}_{\text{default}})$ ▷ periodically refresh the admissible evaluation space
  else
    $\mathcal{M}_t \leftarrow \mathcal{M}_{t-1}$
  end if
end for

Algorithm 2 Evaluator-Side Consistency Reward and Rubric Update

Require: Rollout reports $\{y_i^k\}_{i=1,\ldots,B}^{k=1,\ldots,K}$; sampled rubric (H_t, w_t); rubric prompt x_t^{rubric}; previous EMA baseline b_{t-1}; baseline decay β; evaluator scale α; evaluator update period U

Ensure: Updated EMA baseline b_t, consistency statistics $\{C_i\}_{i=1}^{B}$, and optionally updated shared actor parameters θ

▷ *Phase 1: Build a batch-level consistency signal*

for each query q_i **do**

  Extract rollout conclusions $\{c_i^k\}_{k=1}^{K}$ from $\{y_i^k\}_{k=1}^{K}$ ▷ measure agreement at the conclusion level rather than at the token level

  Compute empirical consistency

$$C_i \leftarrow \textsc{PairwiseConsistency}(\{c_i^k\}_{k=1}^{K})$$

end for

Compute evaluator-side raw signal

$$R_{\text{eval}} \leftarrow \frac{1}{B}\sum_{i=1}^{B} C_i$$

▷ the raw signal favors rubrics under which solver rollouts are more reproducible

▷ *Phase 2: Center the signal with an EMA baseline*

Compute centered evaluator-side signal

$$\hat{A}_{\text{eval}} \leftarrow R_{\text{eval}} - b_{t-1}$$

Update EMA baseline

$$b_t \leftarrow \beta\, b_{t-1} + (1-\beta)\, R_{\text{eval}}$$

▷ the EMA baseline stabilizes REINFORCE updates without changing the raw preference

▷ *Phase 3: Update the evaluator role*

if $t \bmod U = 0$ **and** evaluator updates are enabled **then**

  Update shared actor parameters θ using the evaluator-side REINFORCE objective

$$\theta \leftarrow \theta + \eta\,\alpha\,\hat{A}_{\text{eval}}\,\nabla_\theta \log \pi_e(H_t, w_t \mid x_t^{\text{rubric}}; \theta)$$

▷ rubrics that induce more reproducible solver behavior become more likely

end if

F 详细证明

本附录提供了第 4 节所述结果的推导。全文令 $\theta \in \mathbb{R}^d$ 表示共享参与者参数，并令

$$\widetilde{\mathcal{L}}(\theta) = \mathcal{L}_s(\theta) + \alpha\mathcal{L}_e(\theta) + \lambda\,\mathrm{KL}(\pi_\theta \,\|\, \pi_{\text{ref}})$$

表示 KL 正则化的替代势。

F.1 命题 1 的推导

回顾一下，$\Gamma_q := (H_q, w_q)$ 表示评估者为查询 q 选择的评分标准。假设求解器在相同的查询、证据上下文和评分标准下生成了 K 份报告。设 $c_{ij}(\Gamma_q)$，$1 \le i < j \le K$，表示报告 r_i 和 r_j 在评分标准 Γ_q 下的成对一致性得分，并设

$$M = \binom{K}{2}$$

为报告对的数量。在当前实现中，经验一致性得分定义为

$$C(\Gamma_q) = \frac{1}{M}\sum_{1\le i<j\le K} c_{ij}(\Gamma_q).$$

评估者侧的原始辅助信号为

$$R_e(\Gamma_q) = C(\Gamma_q).$$

设

$$\mu_C(\Gamma_q) := \mathbb{E}[c_{ij}(\Gamma_q)]$$

表示在评分标准 Γ_q 下一致性度量的总体均值。则

$$\mathbb{E}[R_e(\Gamma_q)] = \mathbb{E}[C(\Gamma_q)] = \mathbb{E}\left[\frac{1}{M}\sum_{1\le i<j\le K} c_{ij}(\Gamma_q)\right].$$

根据期望的线性性质，

$$\mathbb{E}[R_e(\Gamma_q)] = \frac{1}{M}\sum_{1\le i<j\le K}\mathbb{E}[c_{ij}(\Gamma_q)] = \frac{1}{M}\sum_{1\le i<j\le K}\mu_C(\Gamma_q) = \mu_C(\Gamma_q).$$

对诱导的评分标准分布取期望，得到

$$\mathbb{E}[R_e(\Gamma_q)] = \mathbb{E}[\mu_C(\Gamma_q)].$$

这证明了命题 1。

在实现中，评估器更新使用中心化信号

$$A_e(\Gamma_q) = R_e(\Gamma_q) - b_t,$$

其中 b_t 是指数移动平均基线。由于 b_t 与当前采样的评分动作无关，减去它可以在不改变 $R_e(\Gamma_q)$ 所编码的底层期望偏好的情况下降低梯度方差。

F.2 命题 2 的推导

共享行动者更新为

$$\theta' = \theta_t - \eta_s\hat{g}_s(\theta_t), \qquad \theta_{t+1} = \theta' - \eta_e\hat{g}_e(\theta').$$

因此，

$$\theta_{t+1} - \theta_t = -\eta_s\hat{g}_s(\theta_t) - \eta_e\hat{g}_e(\theta').$$

对 θ_t 取条件期望可得

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s\mathbb{E}[\hat{g}_s(\theta_t) \mid \theta_t] - \eta_e\mathbb{E}[\hat{g}_e(\theta') \mid \theta_t].$$

在假设（A2）下，

$$\mathbb{E}[\hat{g}_s(\theta_t) \mid \theta_t] \approx \nabla\mathcal{L}_s(\theta_t), \qquad \mathbb{E}[\hat{g}_e(\theta') \mid \theta_t] \approx \nabla\mathcal{L}_e(\theta').$$

因此，

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] \approx -\eta_s\nabla\mathcal{L}_s(\theta_t) - \eta_e\nabla\mathcal{L}_e(\theta').$$

By Assumption (A1), $\mathcal{L}_e$ is L-smooth, so

$$\|\nabla\mathcal{L}_e(\theta') - \nabla\mathcal{L}_e(\theta_t)\| \le L\|\theta' - \theta_t\|.$$

从第一次更新起，

$$\theta' - \theta_t = -\eta_s\hat{g}_s(\theta_t),$$

and by Assumption (A3), $\|\hat{g}_s(\theta_t)\| \le G$. Therefore,

$$\|\theta' - \theta_t\| \le \eta_s G,$$

这意味着

$$\|\nabla\mathcal{L}_e(\theta') - \nabla\mathcal{L}_e(\theta_t)\| \le L\eta_s G.$$

因此我们可以写出

$$\nabla\mathcal{L}_e(\theta') = \nabla\mathcal{L}_e(\theta_t) + \delta_t, \qquad \|\delta_t\| = O(\eta_s).$$

将其代入期望更新，得到

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \mathcal{L}_s(\theta_t) - \eta_e \nabla \mathcal{L}_e(\theta_t) - \eta_e \delta_t.$$

Using $\alpha = \eta_e / \eta_s$, we obtain

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \big(\mathcal{L}_s(\theta_t) + \alpha \mathcal{L}_e(\theta_t) \big) + \rho_t^{(0)},$$

其中

$$\rho_t^{(0)} = -\eta_e \delta_t, \qquad \|\rho_t^{(0)}\| = O(\eta_s^2).$$

加入针对固定参考策略 π_{ref} 的 KL 正则化后，我们定义

$$\widetilde{\mathcal{L}}(\theta) = \mathcal{L}_s(\theta) + \alpha \mathcal{L}_e(\theta) + \lambda \, \mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}) \, .$$

同样的论证随后给出

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \widetilde{\mathcal{L}}(\theta_t) + \rho_t, \qquad \|\rho_t\| = O(\eta_s^2),$$

这证明了命题 2。

F.3 总结

上述推导确立了正文中使用的两个事实：

1. 原始的评估器侧一致性信号估计了在重复求解器展开下由评分标准诱导的期望可复现性；2. 共享参数的求解器-评估器交替更新在一阶近似下遵循 KL 正则化的代理下降方向，仅因顺序更新而产生高阶残差。

G 实验

G.1 训练细节

表 2：SCORE 中的主要训练超参数。

Category	Configuration
Learning rate	1×10^{-6}
Warmup	5 steps, warmup ratio 0.05
Training Data Size	255 for Qwen2.5-7B and 2,380 for Llama-3.1-8B
KL regularization	Coefficient 0.05, low-variance KL, applied in actor loss only
Batch size	Training batch size 64; PPO mini-batch size 32; PPO micro-batch size 4 per GPU
Rollouts	Solver rollout number 8; evaluator rollout number 1
Context and generation limits	Maximum input length 8192; maximum response length 4096; evaluator maximum generation length 2048
Interaction budget	Maximum assistant turns 10
Evaluator update	EMA baseline decay 0.7; update frequency every step; temperature 0.8
Meta-harness	Refresh frequency every 5 training steps
Training retrieval	Local static retrieval backend use wikidata
Compute	Single-node training with 4 H800, 100 CPU cores, and 1 TB host memory
Systems optimization	FSDP, Ray orchestration, gradient checkpointing, parameter offloading, optimizer offloading
Training schedule	50-120 steps

G.2 实验结果

DeepResearchEval 的结果 我们还在 DeepResearchEval [43] 上评估了搜索智能体，其中评估标准针对每个具体查询进行了定制。表 3 所示结果有力地印证了我们之前的观察：基础 GRPO 方法性能严重下降，而我们的框架在所有维度上均取得最高分，包括极具挑战性的查询特定元指标。

表 3： 集成 LLaMA-3.1 和 Qwen2.5 的 open-deep-research 与 gpt-researcher 在 DeepResearchEval 上的性能。Cove：覆盖的广度、深度和相关性。**Insight**：分析的深度、原创性、逻辑性和价值。**IF**：满足所有要求的准确率。Clar：清晰度、流畅性、结构和易理解性。Meta：为每个任务动态生成的查询特定元评估维度。

模型与智能体	Cove	Insight	IF.	Clar	Meta
Agent: Open-deep-research (ReAct Paradigm)					
Llama-3.1-8B-Inst.	3.7	3.2	4.5	5.1	2.6
+ GRPO	1.2 (-2.5)	1.3 (-1.9)	1.4 (-3.1)	1.8 (-3.3)	1.0 (-1.6)
+ DPO	4.3 (+0.6)	4.3 (+1.1)	4.7 (+0.2)	5.2 (+0.1)	3.5 (+0.9)
+ SCORE (Ours)	4.3 (+0.6)	4.4 (+1.2)	4.7 (+0.2)	5.2 (+0.1)	3.5 (+0.9)
Qwen2.5-7B-Inst.	4.4	3.9	5.2	5.4	3.2
+ GRPO	4.4 (+0.0)	4.0 (+0.1)	5.2 (+0.0)	5.4 (+0.0)	3.3 (+0.1)
+ DPO	4.6 (+0.2)	4.1 (+0.2)	5.6 (+0.4)	5.8 (+0.4)	3.4 (+0.2)
+ SCORE (Ours)	6.3 (+1.9)	5.9 (+2.0)	7.0 (+1.8)	6.6 (+1.2)	5.0 (+1.8)
Agent: gpt-researcher (Plan-and-Execute Paradigm)					
Llama-3.1-8B-Inst.	3.9	3.5	4.8	5.6	2.8
+ GRPO	2.5 (-1.4)	2.8 (-0.7)	2.7 (-2.1)	3.7 (-1.9)	2.0 (-0.8)
+ DPO	2.4 (-1.5)	2.8 (-0.7)	2.6 (-2.2)	3.5 (-2.1)	2.0 (-0.8)
+ SCORE (Ours)	5.9 (+2.0)	5.6 (+2.1)	6.9 (+2.1)	6.9 (+1.3)	4.6 (+1.8)
Qwen2.5-7B-Inst.	4.5	4.1	5.7	5.9	3.4
+ GRPO	4.6 (+0.1)	4.1 (+0.0)	5.6 (-0.1)	5.8 (-0.1)	3.4 (+0.0)
+ DPO	6.1 (+1.6)	5.8 (+1.7)	6.8 (+1.1)	6.5 (+0.6)	4.8 (+1.4)
+ SCORE (Ours)	6.5 (+2.0)	6.0 (+1.9)	7.5 (+1.8)	7.0 (+1.1)	5.1 (+1.7)

有趣的是，尽管在 DeepResearchBench 基准上通用可读性有所下降，但我们的方法在 DeepResearchEval 的"清晰度"指标上取得了持续提升。这种差异支持了我们的权衡假设，证实生成的文本以对话简洁性换取高度结构化和分析清晰的呈现方式，这在评估复杂的查询特定研究解释时获得高度奖励。

训练数据规模 我们在 SCORE 中应用了不同的训练数据规模。虽然共同进化的评估器成功地稳定了较小数据集上的训练，但增加数据规模会加速模型崩溃。原因在于求解器和评估器都依赖于本能奖励，而不是严格可验证的外部真实标准。在训练数据有限的情况下，这种精确的内部反馈有效地强化了模型的正确先验，并提供了必要的密集约束。然而，随着数据集规模显著增加，求解器逐渐发现评估器的盲点，并对这些代理奖励进行过度优化。这种动态不可避免地迫使优化过程饱和，随着求解器利用代理反馈的分布极限，事实准确率下降。如图 5 所示，求解器熵在大规模下过早崩溃，随后训练奖励呈现经典的先升后降轨迹。这一观察结果与最近关于无监督强化学习的理论发现完全一致，证实了内在奖励机制在暴露于大量数据时最终会放大自我强化的幻觉 [17]。

(a) Solver Entropy (b) Evaluator Reward (c) Solver Reward

图 5：训练细节

Case Analysis

General Evaluation Rubric: [factuality, insight, coherence, complexity]

User Query 1: What does Texas Penal Code Chapter 9.42 specifically permit, and under what conditions can deadly force be used to protect property?
Report: Texas law permits deadly force to protect property when the force is deemed reasonable, especially in cases involving theft or unlawful intrusion. The rule is often understood as allowing force when property cannot otherwise be protected. In practice, this reflects a broad legal recognition of property rights and self-defense principles. Courts have generally supported the idea that homeowners ...

Static evaluation: [factuality: 7, insight: 2, coherence: 8, complexity: 7]
Adaptive evaluation: [factuality: 3, citation_effectiveness: 2, coherence: 8, completeness: 3]
Reason: The report is well-written, but it does not ground its claims in the statutory text and provide valid citation. Specifically, it omits the exact conditions enumerated in §9.42, such as the nighttime requirement and the "no other means" clause, leading to an overly permissive interpretation.

User Query 2: What are the main debates around replacing human workers with AI assistants in knowledge-intensive jobs?
Report: AI can improve efficiency, reduce repetitive work, and increase access to expertise across organizations. They may also lower costs and support faster decision-making in research, operations, and customer service. At the same time, concerns remain about job displacement, deskilling, accountability, and over-reliance on automated systems. Overall, most commentators agree that AI will reshape knowledge work, and organizations should gradually integrate these tools while monitoring their impact ...

Static evaluation: [factuality: 7, insight: 2, coherence: 8, complexity: 2]
Adaptive evaluation: [diversity: 3, fairness: 4, coherence: 8, complexity: 4]
Reason: The report is coherent and thoughtful, but it presents only a limited set of viewpoints and does not sufficiently cover the competing perspectives.

图 6：不同查询的自适应评估与静态评估。

案例研究 图 6 展示了静态评估和自适应评估在评分行为上的鲜明对比。在静态评分标准下，求解器成功地利用了通用的"连贯性"度量，通过生成表面流畅的散文，在两种场景中都获得了接近完美的 8 分。然而，这种高结构分数掩盖了实质性内容的关键遗漏。当应用我们的自适应框架时，它会立即综合查询特定的约束来审问文本。在法律场景中，动态评估标准关注报告是否将其主张建立在具体的法定前提之上。同样，在讨论场景中，度量标准暴露了报告未能综合对立观点的问题，尽管其语气看似深思熟虑。这种动态评估有效地防止了模型优先考虑优雅的措辞来获得高分，而忽视事实准确性或其他关键维度要求。

H 提示词

```
You will be provided with a reference and some statements. Please determine
    whether each statement is 'supported', 'unsupported', or 'unknown' with
    respect to the reference. Please note:
First, assess whether the reference contains any valid content. If the
    reference contains no valid information, such as a 'page not found'
    message, then all statements should be considered 'unknown'.
If the reference is valid, for a given statement: if the facts or data it
    contains can be found entirely or partially within the reference, it is
    considered 'supported' (data accepts rounding); if all facts and data in
    the statement cannot be found in the reference, it is considered '
    unsupported'.

You should return the result in a JSON list format, where each item in the
    list contains the statement's index and the judgment result, for example:
[
    {{
        "idx": 1,
        "result": "supported"
    }},
    {{
        "idx": 2,
        "result": "unsupported"
    }}
]

Below are the reference and statements:
<reference>
{reference}
</reference>

<statements>
{statements}
</statements>

Begin the assessment now. Output only the JSON list, without any
    conversational text or explanations.
```

Listing 1: Prompt for citation checking.

```
<system_role>
You are an experienced research article evaluation expert. Your task is to
    evaluate a deep research article based on the provided `<task>` across
    four dimensions: Comprehensiveness, Insight, Instruction Following, and
    Readability.
</system_role>

<task>
"{task_prompt}"
</task>

<instruction>
Please deeply analyze the task and evaluate the research report based on the
    following specific criteria for each dimension:

1. Comprehensiveness (Information Coverage)
- Evaluate the breadth, depth, and relevance of information coverage.
- Ensure the article covers key information areas, perspectives, and depths
    necessary to achieve comprehensive market or situational analysis.
- Check if it thoroughly examines all relevant factors (e.g., technical,
    economic, social, geographical) and avoids omitting critical facets of the
     topic.

2. Insight (Analytical Depth)
- Evaluate the depth, originality, logic, and value of the analysis and
    conclusions.
```

```
- Assess whether the analysis goes beyond a superficial listing of factors to
     uncover core drivers, subtle mechanisms, and second-order effects.
- Examine if the conclusions and recommendations are logically derived from
    the analysis, offer unique perspectives, and provide forward-looking or
    actionable strategic value.

3. Instruction Following (Responsiveness & Relevance)
- Evaluate whether the report accurately, completely, and directly responds
    to all specific instructions, questions, and core objectives within the `<
    task>`.
- Check for strict adherence to scope limitations (e.g., specific geography,
    time, or subject constraints) without significant deviation into
    irrelevant topics.

4. Readability (Presentation Quality)
- Evaluate the clarity of structure, fluency of language, effectiveness of
    data presentation, and overall ease of understanding.
- Assess structural logic (clear framework and logical flow), language
    precision (fluent, grammatically correct, and appropriate use of
    terminology), and cohesive paragraph transitions.
- Check the clarity and accuracy of data presentation, including the
    effective use of visualizations (charts, tables) and the highlighting of
    key findings to reduce reader fatigue.
</instruction>
```

Listing 2: Prompt for writing quality evaluation

```
[System Prompt]
You are an external researcher. Adjust the evaluator harness based on recent
    training statistics. Do not choose the evaluator's final dimensions or
    weights directly.
To request search from the evaluator, set require_search=true in
    process_constraints, and explicitly state in notes that the evaluator
    should use the search tool by outputting <search>query</search> tags.

[User Prompt]
Task: Generate the next evaluator harness H_t.
Goal: Update evaluator constraints without directly choosing concrete
    dimensions or weights for the evaluator. You may adjust
    dimension_constraints and process_constraints and dimension_pool.

## Statistics:
{Training Statistics JSON}


## Output Format:
{
  "dimension_constraints": {
    "k": number of dimensions,
    "must_include": [list of required dimension names],
    "weight_min": minimum weight for a dimension,
    "weight_max": maximum weight for a dimension
  },
  "process_constraints": {
    "require_gap_analysis": true/false,
    "require_verification": true/false,
    "citation_cross_check": true/false,
    "min_number_of_tool_call": n
  },
  "notes": "additional process compliance instructions"
}
```

Listing 3: Prompt for meta-harness

```
[System Prompt]
You are a professional evaluator for deep research reports.
You will receive one query and multiple candidate reports generated by the
    solver.
```

```
Your job is to evaluate every report with a search-assisted review process.

You must complete 4 stages:
1. {must_include} is always required as the primary scoring dimension. Select
     exactly {k-1} additional dimensions from the remaining pool and assign
    weights, each additional dimension's weight must be within [{Weight Min},
    {Weight Max}]
2. Verify key claims and citations in each report using the available search
    tool.
3. Perform independent research from the original query to identify important
     gaps not covered by each report.
4. Score each report on every selected dimension using the evidence you found
    .

## Tool-use protocol:
- When you need external evidence, you can use search tools with search query
     in this format:
  <search>your search query here</search>
- The content inside <search> must be only the query text. Do not output JSON
    , function-call syntax, XML attributes, bullet lists, or multiple queries.
- After a search, the system will return retrieved evidence inside <
    observation>...</observation>.
- You may issue multiple search queries as needed before producing the final
    XML.

Return the final result in the following XML format ONLY AFTER you finish
    your search and evaluation:
<dimensions>
  <dim name="[Base Dimension]"/>
  <dim name="[Selected Dimension]" weight="[Weight]"/>
</dimensions>
<report_evaluations>
  <report id="[Report ID]">
    <verification>your verification process here</verification>
    <gap_analysis>your gap analysis here</gap_analysis>
    <scores>
      <score dim="[Base Dimension]" value="[Score]" reason="..."/>
      <score dim="[Selected Dimension]" value="[Score]" reason="..."/>
    </scores>
  </report>
</report_evaluations>
```

Listing 4: Prompt for evaluator

```
[System Prompt]
You are a professional deep researcher. Solve the user's request by
    iteratively searching public information and synthesizing evidence.

## Rules:
1. Think inside <think> and </think> before each action and after each new
    search result.
2. When external evidence is needed, emit exactly one standalone search query
    . Use the exact format <search>your query here</search> to issue a search.
3. The content inside <search> must be only the query text. Do not output
    JSON, tool names, XML attributes, bullet lists, or multiple queries in one
     search block.
4. Prefer natural web search queries: start broad, then narrow to
    verification, recent developments, entity-specific facts, or missing
    perspectives.
5. Prefer multiple sequential searches over one overloaded query.
6. Use search to verify important claims, gather evidence, and close
    information gaps. Stop searching once you can answer confidently.
7. The environment will return search results between <observation> and </
    observation>.
8. When you have enough evidence, write the final response only inside <
    answer> and </answer>.
9. In the final answer, synthesize the evidence, note uncertainty when
    necessary, and include source mentions or a short Sources section.
```

清单 5：求解器的提示词